

Planning-Aligned Pretraining of BEV Representations with Sparse Action-Conditioned Targets for End-to-End Autonomous Driving

Anonymous submission

Abstract

End-to-end autonomous driving relies on a shared bird’s-eye-view (BEV) representation for perception, prediction, and planning, yet the BEV encoder is typically optimized only through heterogeneous downstream tasks. This leaves no dedicated stage for organizing the representation around geometry relevant to candidate actions. We introduce PAVER, Planning-Aligned BEV Encoder Pretraining, which rasterizes a LiDAR sweep into sparse free, occupied, and unknown cells. Rule-based candidate ego motions are rolled out and queried laterally across the vehicle width; the fractions of queries falling on measured returns and on unobserved cells form sparse action targets. To limit direct copying from the supervised regions, PAVER replaces action-corridor features with a shared learnable mask token and predicts these targets from masked camera-derived BEV features conditioned on the corresponding action state. The framework requires no driving-task annotations or learned teacher and introduces only 10K trainable auxiliary parameters. After pretraining, the target builder and prediction components are discarded and only the BEV encoder is transferred, leaving the downstream architecture and inference pipeline unchanged. On nuScenes, PAVER lowers VAD-Tiny’s average collision rate from 0.51% to 0.19% and its planning L2 from 0.66 to 0.60, while improving motion ADE from 0.91 to 0.80 and map mAP from 0.42 to 0.44. Planning L2 improves on VAD-Base and GenAD as well, although the collision gain does not carry over to VAD-Base.

Introduction

Camera-based end-to-end driving increasingly organizes detection, map prediction, motion prediction, and planning around a shared bird’s-eye-view (BEV) representation (Hu et al. 2022, 2023; Jiang et al. 2023; Sun et al. 2024; Zheng et al. 2024; Liao et al. 2025). Training these systems remains costly: the BEV encoder must acquire cross-view geometry, semantics, interactions, and planning-relevant structure through long, coupled downstream schedules (Hu et al. 2023; Jiang et al. 2023). Standard image-pretrained backbones initialize the image feature extractor, but not the shared BEV construction or its planning-specific organization (Deng et al.

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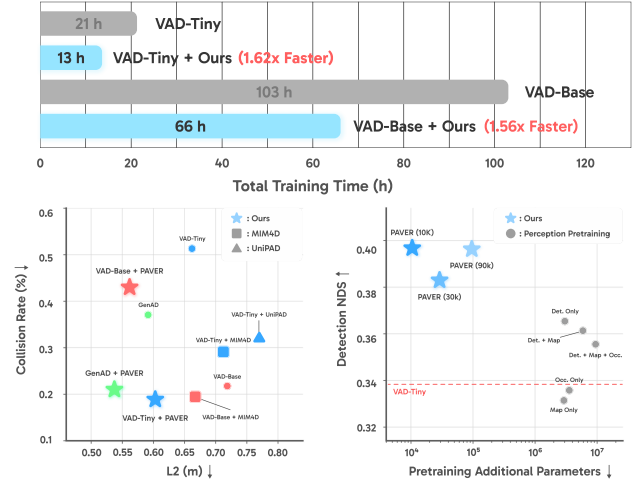


Figure 1: PAVER reaches the baseline planning accuracy in less total training time, including its own pretraining, measured on 4 NVIDIA RTX 5090 GPUs, and it does so with a smaller planning error, a lower collision rate, and a higher detection score than pretraining that adds orders of magnitude more parameters.

2009; He et al. 2016; Li et al. 2022); both must still emerge through downstream optimization.

Existing systems commonly manage this burden through task-dependent staging: perception, tracking, or mapping is optimized first and then used to initialize a larger end-to-end model. Although effective, this approach still depends on dense annotations, complex task heads, and objectives tied to a particular decoder design (Hu et al. 2022, 2023; Jiang et al. 2023; Weng et al. 2024; Doll et al. 2024; Tang et al. 2025; Li and Cui 2025). We take the alternative position that BEV initialization need not reproduce a complex downstream task, provided that pretraining supervises the representation where planning will query it.

Driving-specific pretraining methods provide relevant geometry, yet representative objectives remain centered on reconstructing scene content (Yang et al. 2024a; Zou et al. 2025; Zhang et al. 2025). UniPAD reconstructs continuous 3D geometry and appearance, MIM4D reconstructs masked multi-view video, and VisionPAD uses rendering and tem-

poral consistency. As summarized in Fig. 2, these methods supervise general spatial support rather than the regions most relevant to planning. Planning instead depends on which observed regions constrain candidate actions (Hu et al. 2023). This mismatch motivates supervising the BEV regions used for planning.

We introduce **PAVER**, *Planning-Aligned BEV Encoder Pretraining*, which pretrains an otherwise unchanged BEV encoder. Following occupancy-grid mapping (Elfes 1989), a LiDAR sweep is reduced to sparse free, occupied, and unknown evidence. Rule-based actions index supervised BEV regions, where samples across the vehicle footprint form sparse action targets for *risk* and *unknown* evidence. PAVER replaces these corridor features with a shared mask token and predicts the corresponding sparse action targets conditioned on the action state. This directs BEV pretraining toward evidence that constrains candidate actions.

PAVER requires neither task-specific annotations, a learned teacher, nor dense 3D scene reconstruction. The pretraining-only PAVER head adds 10K parameters and is removed before downstream training; only the BEV encoder is transferred, preserving the downstream architecture and inference pipeline. The transfer protocol is detailed in the Method section, and the auxiliary capacity is reported in Table 7. With this lightweight supervision and shorter total training time, PAVER matches or improves the corresponding baseline on most evaluated multi-task metrics (Tables 9 and 1). Collision behavior nevertheless varies across architectures (Table 4).

Our contributions are:

- We formulate *planning-aligned pretraining of the BEV encoder* as a lightweight alternative to learning shared BEV representations only through long, task-coupled schedules.
- We instantiate this formulation with sparse action targets from LiDAR risk and unknown evidence and a masked, action-conditioned BEV branch, without RGB/depth reconstruction, future prediction, or a learned teacher.
- We transfer only the pretrained BEV encoder and remove the 10K-parameter PAVER head after pretraining. Across multiple backbones, we evaluate downstream multi-task performance and analyze how decoder design affects planning safety.

Related Work

Shared representations and initialization in end-to-end driving. End-to-end driving systems couple perception, prediction, and planning through a shared spatial representation, ranging from dense BEV grids (Li et al. 2022, 2023; Huang et al. 2023) to vectorized or sparse scene elements (Hu et al. 2023; Jiang et al. 2023; Sun et al. 2024; Zheng et al. 2024; Li and Cui 2025). Initialization is commonly obtained from downstream supervision: ST-P3, UniAD, and VAD stage perception-oriented training before the broader task stack, PARA-Drive redesigns task connectivity for joint optimization, and later systems vary the decoder itself (Hu et al. 2022, 2023; Jiang et al. 2023; Weng et al. 2024; Doll

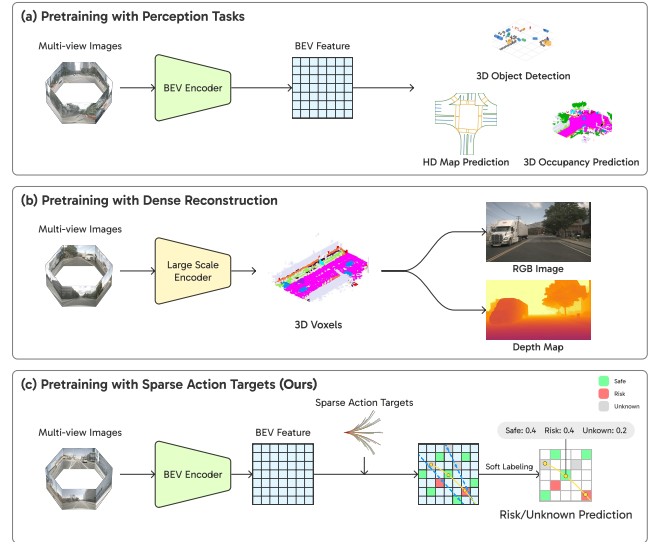


Figure 2: Driving pretraining paradigms. Existing work supervises pretraining with (a) perception labels or (b) dense reconstruction; (c) PAVER introduces sparse action targets.

et al. 2024; Tang et al. 2025; Li and Cui 2025). The representation is therefore shaped by model-specific annotations and decoders. PAVER separates initialization from that optimization, targeting architectures that retain a dense, spatially indexed BEV representation and leaving their task heads unchanged.

Driving-specific 3D pretraining. Reconstruction methods learn transferable geometry from masked or projected scene content. UniPAD reconstructs continuous 3D structure and appearance, MIM4D extends reconstruction to multi-view video, and VisionPAD uses rendering and temporal consistency to learn 3D-aware features (Yang et al. 2024a; Zou et al. 2025; Zhang et al. 2025). These methods lift image features into a 3D volume and supervise it densely, so the pretrained weights that transfer are the image backbone rather than the shared BEV encoder. BEV-MAE and Occupancy-MAE similarly reconstruct masked spatial structure for LiDAR representations (Lin et al. 2024; Min et al. 2022). A separate line distills LiDAR knowledge into camera models through feature or label alignment (Chen et al. 2023; Wang et al. 2023; Zhou et al. 2023; Kim et al. 2024; Hinton, Vinyals, and Dean 2015; Vapnik and Izmailov 2015). PAVER pretrains the BEV encoder itself, uses no learned LiDAR encoder, and performs no feature matching: deterministic free, occupied, and unknown evidence is aggregated into sparse action targets.

Predictive world models. World-model pretraining learns representations by predicting how scenes evolve. DriveWorld models spatiotemporal latent dynamics, while ViDAR predicts future geometric observations from camera history (Min et al. 2024; Yang et al. 2024b). LAW, DriveX, and DLWM connect future scene or latent prediction more directly to planning (Li et al. 2025; Shi et al. 2025; Zhu et al. 2026). Such objectives provide temporal and dynamic su-

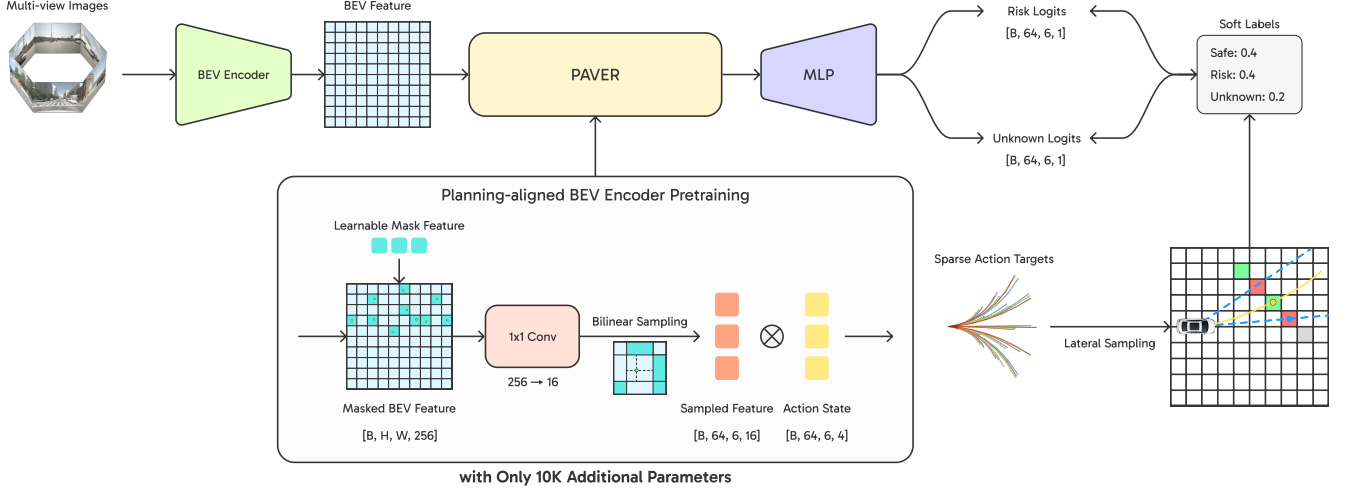


Figure 3: PAVER overview. PAVER constructs sparse action targets for risk and unknown evidence, masks their associated BEV regions, and predicts the targets from the remaining context. The PAVER head is removed before downstream training.

pervision, but they depend on future observations, future geometry, or learned transition targets. PAVER addresses a complementary setting: it uses only the current synchronized sweep and does not forecast future agents or future scene states. Its sparse action targets summarize present geometric evidence rather than future world state.

Action and planning signals in representation learning.

Actions have appeared in driving pretraining as labels, conditioning variables, policy targets, or guides for latent prediction (Zhang, Peng, and Zhou 2022; Xiao et al. 2021; Wang et al. 2026; Chen et al. 2020; Li et al. 2025; Zhu et al. 2026). PAVER instead uses rule-based actions as spatial indices for sparse action targets that aggregate current free, occupied, and unknown evidence, so the encoder never reproduces an expert action, a demonstrated trajectory, or a future latent state.

Method

Overview

PAVER pretrains the BEV encoder with sparse action targets constructed from LiDAR evidence along rule-based ego motions. Multi-view images I are processed by the same BEV encoder used for downstream end-to-end training (Lin et al. 2017; Li et al. 2022; Jiang et al. 2023), producing $F = f_{\text{bev}}(I) \in \mathbb{R}^{B \times C \times H \times W}$. LiDAR measurements are used only to construct pretraining targets. Detection, mapping, motion-prediction, and planning annotations are not used, and the corresponding task decoders are disabled during pretraining. All parameters of f_{bev} are optimized by the pretraining objective and transferred to downstream training (Fig. 4). The action generator, LiDAR target builder, and the **PAVER head** (Fig. 4), which comprises the learnable mask token, pointwise projector, and prediction MLP, are used only during pretraining.

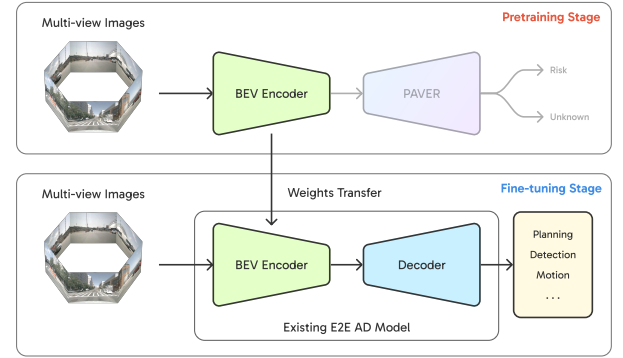


Figure 4: Two-stage training protocol. Only the pretrained BEV encoder is transferred, and the existing end-to-end model is fine-tuned unchanged.

Candidate Ego Motions

Candidate motions define the spatial locations at which supervision is constructed. They are generated from rule-based controls rather than expert trajectories, ground-truth future motions, or learned action proposals.

Finite sets of longitudinal accelerations and yaw rates are combined into rule-based control families, tiled across action slots, and independently perturbed during training. Each control is rolled out from the current ego speed over T horizons, giving the ego-frame state $q_{s,t}^{\text{ego}} = (x_{s,t}^{\text{ego}}, y_{s,t}^{\text{ego}}, \psi_{s,t}^{\text{ego}}, v_{s,t})$ for action slot s and horizon t , where ψ is the yaw heading. Rollout positions and headings are transformed into the LiDAR/BEV indexing frame, yielding $\bar{q}_{s,t}^{\text{bev}} = (\bar{x}_{s,t}, \bar{y}_{s,t}, \bar{\psi}_{s,t}, v_{s,t})$. The transformed state is used for LiDAR target construction and BEV feature sampling, while the PAVER head is conditioned on the original ego-frame state.

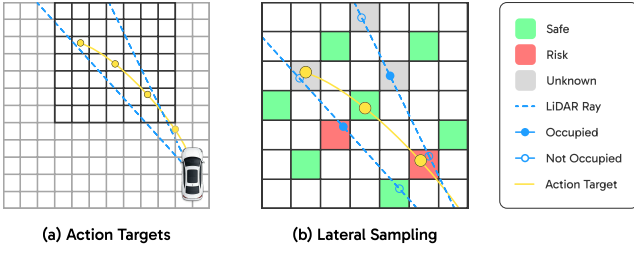


Figure 5: Sparse action-target construction. LiDAR rays provide Risk and Unknown targets for action-indexed queries; “Safe” denotes Free evidence.

Sparse LiDAR Evidence

Following occupancy-grid mapping (Elfes 1989), we rasterize the LiDAR measurements into a BEV evidence map $E \in \{\text{UNKNOWN}, \text{FREE}, \text{OCCUPIED}\}^{B \times H \times W}$ without a learned LiDAR encoder. Cells start as UNKNOWN; samples along each open ray segment are marked FREE (Amanatides and Woo 1987) and the measured endpoint then overrides its cell as OCCUPIED. The raster therefore separates observed free space, measured returns, and regions the rays never support. It only builds pretraining targets and is never predicted by the camera model as a dense occupancy map.

Sparse Action Targets

At each candidate state $\bar{q}_{s,t}^{\text{bev}}$ we query K locations laterally across the vehicle width, and $\mathcal{V}_{s,t}$ denotes the queries that stay in range. The two targets are the fractions of those queries that fall on measured endpoints and on cells without supporting ray evidence:

$$r_{s,t} = \frac{|\{k \in \mathcal{V}_{s,t} : E_k = \text{OCCUPIED}\}|}{|\mathcal{V}_{s,t}|}, \quad (1)$$

$$u_{s,t} = \frac{|\{k \in \mathcal{V}_{s,t} : E_k = \text{UNKNOWN}\}|}{|\mathcal{V}_{s,t}|}, \quad (2)$$

where E_k is the evidence value at the k -th query position.

The two targets are independent continuous ratios rather than mutually exclusive classes, and they summarize only what the sensor observed at the input time: risk marks a queried location the sweep reports as occupied, unknown marks one the sweep does not support at all, and neither is a future collision probability, a calibrated uncertainty, or a forecast of agent motion. Candidate motions supply the coordinates at which this evidence is aggregated and are never prediction targets. This is what we mean by *planning-aligned* supervision: targets are placed along rule-based ego motions instead of densely over the BEV raster. The lateral offsets, the range test, and the resulting tensor shapes are listed in the Supplementary Material.

Action-Corridor Feature Masking

The target-associated cells may contain local BEV features that directly reflect the same geometric evidence used to construct the targets. To limit the access the PAVER head has to these vectors, we build a binary action-corridor mask $M \in$

$\{0, 1\}^{B \times 1 \times H \times W}$ from the union of all rasterized lateral-query cells, and replace the selected cells of a temporary copy with a shared learnable mask token $m \in \mathbb{R}^{1 \times C \times 1 \times 1}$:

$$F^{\text{mask}} = (1 - M) \odot F + M \odot m. \quad (3)$$

Unselected cells retain their original features and the original representation F is left unchanged. This removes the original local feature vectors at the selected cells from the input of the PAVER head; it does not guarantee the use of neighboring cells or remove information already encoded before masking.

Action-Conditioned Prediction

A pointwise 1×1 projector ϕ maps the masked features to $Z = \phi(F^{\text{mask}})$, transforming channels independently at each cell without spatial aggregation. For each action slot and horizon, one projected feature $z_{s,t}$ is bilinearly sampled at the transformed action center $(\bar{x}_{s,t}, \bar{y}_{s,t})$ (Jaderberg et al. 2015), concatenated with the ego-frame action state, and mapped by the prediction MLP g to two independent logits:

$$(\hat{r}_{s,t}, \hat{u}_{s,t}) = g([z_{s,t}; q_{s,t}^{\text{ego}}]). \quad (4)$$

Because the lateral-offset set contains zero, the action-center cell is itself masked. The action state specifies which candidate position, heading, and velocity are associated with the sampled feature.

The LiDAR and camera branches use different spatial samples. The K lateral LiDAR queries are aggregated into one target pair, whereas the camera branch samples one center feature per action-horizon entry. The predictor therefore estimates one risk and unknown pair per entry rather than classifying individual lateral cells.

Pretraining Objective

The pretraining objective is

$$\mathcal{L}_{\text{pre}} = \lambda_r \mathcal{L}_r + \lambda_u \mathcal{L}_u, \quad (5)$$

where $\mathcal{L}_r$ and $\mathcal{L}_u$ are logit-based binary cross-entropy losses for the continuous risk and unknown targets, averaged over valid action-horizon entries.

After pretraining, the PAVER head, candidate-motion generator, LiDAR rasterization, and target builder are discarded. Only the pretrained BEV encoder initializes the downstream camera-only model, while its task-specific heads retain their standard initialization. The downstream architecture, training objectives, and inference procedure remain unchanged.

The rollout equations, coordinate transformation, control sets, perturbation parameters, lateral offsets, tensor dimensions, loss weights, optimization settings, and parameter-transfer rules are given in the Method and Reproducibility Details section of the Supplementary Material.

Experiments

Implementation Details

We pretrain on the 28,130 nuScenes training keyframes and evaluate on all 6,019 validation keyframes, using VAD-Tiny (Jiang et al. 2023) unless stated otherwise. Two constraints keep the comparison clean: PAVER uses a single-frame temporal queue, so pretraining cannot exploit a recurrent BEV

Method	Epochs		Auxiliary Tasks		Planning		Motion			Detection		Map
	PT	FT	Pretraining	Fine-tuning	L2 (m)	Col. (%)	ADE	FDE	MR	mAP	NDS	mAP
VAD-Tiny [†]	0	60	None	Det., Map, Mot., Plan.	0.66	0.51	0.91	1.25	0.13	0.23	0.34	0.42
+ Ours	20	30	Actions	Det., Map, Mot., Plan.	0.60	0.19	0.80	1.09	0.12	0.28	0.40	0.44
VAD-Base	0	60	None	Det., Map, Mot., Plan.	0.72	0.22	0.69	0.90	0.09	0.33	0.46	0.48
+ Ours	20	30	Actions	Det., Map, Mot., Plan.	0.56	0.43	0.69	0.90	0.09	0.33	0.45	0.50
GenAD [†]	0	60	None	Det., Map, Mot., Plan.	0.59	0.37	0.90	1.23	0.13	0.19	0.26	0.46
+ Ours	20	30	Actions	Det., Map, Mot., Plan.	0.54	0.21	0.80	1.05	0.11	0.21	0.28	0.44

[†] Reproduced results.

Table 1: Multi-task performance on the nuScenes validation set. Actions denotes PAVER’s sparse action targets. PT and FT denote pretraining and fine-tuning epochs; Det., Mot., and Plan. denote detection, motion prediction, and planning. L2 is the trajectory displacement error and Col. the box collision rate in percent, both averaged over 1, 2, and 3 seconds; ADE, FDE, and MR are the motion average and final displacement errors and miss rate; mAP and NDS are mean average precision and the nuScenes detection score.

state, and LiDAR enters only through the targets, leaving downstream training and inference camera-only.

We pretrain for 20 epochs on 4 GPUs with AdamW (Loshchilov and Hutter 2019) and cosine annealing (Loshchilov and Hutter 2017), lowering the image-backbone learning rate so that pretraining reorganizes the BEV construction rather than rewriting image features; downstream training keeps each model’s original recipe. At transfer we retain only the BEV encoder and reinitialize every task decoder. Each reported row comes from 1 checkpoint selected for joint multi-task balance within its budget, so no row combines the best value of each metric. Remaining settings are in the Supplementary Material.

We evaluate the transferred models on nuScenes (Caesar et al. 2020) using the planning, motion-prediction, detection, and vector-map metrics adopted by VAD (Jiang et al. 2023). Planning is measured by trajectory L2 error and box collision rate, motion prediction by ADE, FDE, and miss rate, detection by mAP and NDS, and mapping by vector-map mAP. Table 1 reports planning metrics averaged over 1, 2, and 3 seconds; per-horizon values are reported separately.

Results with Pseudo-LiDAR

Target Source	Planning		Motion			Detection	Map
	L2 (m) ↓	Col. (%) ↓	ADE	FDE	MR	NDS	mAP
None [†]	0.662	0.513	0.905	1.250	0.135	0.338	0.419
Pseudo-LiDAR	0.563	0.560	0.831	1.101	0.121	0.386	0.439
LiDAR	0.603	0.187	0.803	1.086	0.121	0.397	0.439

[†] Reproduced results.

Table 2: Planning results with pseudo-LiDAR pretraining. Planning and motion metrics are averaged over 1, 2, and 3 seconds. None denotes VAD-Tiny trained from scratch.

PAVER does not require a LiDAR sensor. Where none is available, pseudo-LiDAR from a frozen UniK3D metric-depth estimator (Piccinelli et al. 2025), unprojected per camera (Wang et al. 2019), supplies the same surface endpoints

and ray segments the target builder expects. Table 2 shows the substitution works: estimated depth improves L2, ADE, NDS, and map mAP over scratch, though collision does not follow, so it is a usable but non-equivalent target source.

Method	AP ↑			mAP
	Divider	Ped. Crossing	Boundary	
VAD-Tiny [†]	0.42	0.31	0.46	0.40
+ Ours	0.48	0.34	0.49	0.44
VAD-Base	0.52	0.41	0.51	0.48
+ Ours	0.53	0.44	0.52	0.50
GenAD [†]	0.50	0.38	0.49	0.46
+ Ours	0.47	0.35	0.48	0.44

[†] Reproduced results.

Table 3: Map prediction results on nuScenes. AP and mAP denote average precision per map class and its mean; Ped. Crossing denotes pedestrian crossing.

Quantitative Results on nuScenes

A pretraining head carrying no task annotation is enough to match or improve most downstream metrics on three architectures (Table 1), and vector-map prediction improves for both VAD variants in every map class even though no map annotation is used during pretraining (Table 3). The pre-trained encoder therefore carries structure the map decoder can reuse. GenAD is the exception, so the benefit depends on the decoder as well as the objective.

Displacement and collision do not move together. VAD-Tiny and GenAD improve both at every horizon, whereas VAD-Base improves displacement and worsens collision at every horizon (Table 4), locating the disagreement in the decoder rather than the objective.

Table 5 carries the comparison that motivates the objective: against reconstruction-based pretraining, PAVER reaches lower L2 at every VAD-Tiny horizon with roughly

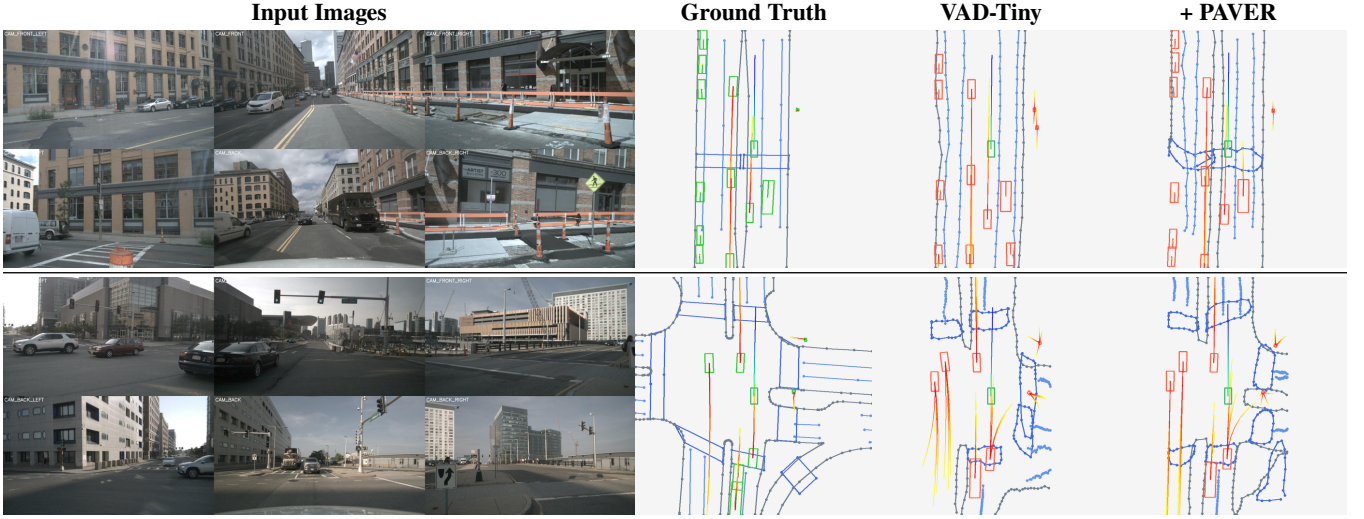


Figure 6: VAD-Tiny qualitative comparisons on nuScenes. More results are in the Supplementary Material.

Method	L2 (m) ↓			Col. (%) ↓		
	1s	2s	3s	1s	2s	3s
VAD-Tiny [†]	0.35	0.63	1.01	0.38	0.45	0.71
+ Ours	0.32	0.58	0.92	0.09	0.15	0.32
VAD-Base	0.41	0.70	1.05	0.07	0.17	0.41
+ Ours	0.30	0.53	0.85	0.26	0.43	0.61
GenAD [†]	0.33	0.56	0.88	0.21	0.34	0.56
+ Ours	0.28	0.51	0.82	0.09	0.18	0.35

[†] Reproduced results.

Table 4: Planning results by horizon on nuScenes, reported at the 1, 2, and 3 second horizons.

three orders of magnitude fewer auxiliary parameters (0.01M against 6.41M and 13.43M). What matters is where supervision is placed, not how much of the scene it reproduces. To our knowledge this is the first driving pretraining objective supervised only at action-indexed BEV locations, without scene reconstruction, a learned teacher, or future prediction. Attributing the BEV features back to the front camera in Fig. 7, PAVER concentrates on objects that constrain the planned motion; more results are in the Supplementary Material.

Closed-Loop Evaluation

Town05 Long and Bench2Drive (Jia et al. 2024) test whether the same initialization survives closed-loop control, where errors accumulate over a route instead of being scored per frame. Table 6 shows the improvement is not confined to open-loop metrics: on UniAD-Tiny the pretrained model raises Driving Score and Route Completion over its baseline under an identical protocol. The benchmark uses 9 routes with one repetition, so it indicates transfer rather than a ranking.

Method	Target	#Params	L2 (m) ↓			Col. (%) ↓		
			1s	2s	3s	1s	2s	3s
VAD-Tiny [†]	—	0.00M	0.46	0.76	1.12	0.21	0.35	0.58
+ UniPAD [‡]	RGB, Depth	6.41M	0.45	0.75	1.11	0.19	0.31	0.46
+ MIM4D [‡]	RGB, Depth	13.43M	0.39	0.69	1.06	0.21	0.28	0.38
+ Ours	Sparse Actions	0.01M	0.32	0.58	0.92	0.09	0.15	0.32
VAD-Base	—	0.00M	0.41	0.70	1.05	0.07	0.17	0.41
+ MIM4D [‡]	RGB, Depth	13.43M	0.36	0.64	1.00	0.08	0.14	0.36
+ Ours	Sparse Actions	0.01M	0.30	0.53	0.85	0.26	0.43	0.61

[†] Reproduced results. [‡] Reported results.

Table 5: Planning performance under different pretraining methods. RGB denotes camera-color reconstruction, and Depth denotes LiDAR-projected sparse metric-depth reconstruction. Sparse Actions denotes PAVER’s sparse action targets. #Params reports temporary auxiliary parameters.

Method	DS ↑	RC ↑	IS ↑
UniAD-Tiny	48.45	60.96	0.85
+ Ours	58.79	79.06	0.79
GenAD [‡]	34.53	—	—
+ Ours	49.07	58.95	0.90

[‡] Reported by GenAD.

Table 6: Closed-loop results on Bench2Drive Town05 Long. DS, RC, and IS denote Driving Score, Route Completion, and Infraction Score, respectively; a dash marks a metric the corresponding source does not report. Closed-loop qualitative results are in the Supplementary Material.

Ablation Study

We next examine which parts of the pretraining objective account for the transfer results and whether sparse supervision affects features outside the action corridors.

Pretraining				Plan.		Mot.	Det.	Map
Det.	Map	Occ. [†]	Act. #Params ↓	L2 (m) ↓	Col. (%) ↓	ADE ↓	NDS ↑	mAP [‡]
			0.00M	0.662	0.513	0.905	0.338	0.419
✓			3.01M	0.515	0.240	0.798	0.365	0.406
	✓		2.91M	0.694	0.333	0.857	0.331	0.391
		✓	3.56M	0.646	0.320	0.905	0.336	0.393
✓	✓		5.92M	0.504	0.190	0.805	0.361	0.397
✓	✓	✓	9.49M	0.577	0.270	0.798	0.356	0.420
		✓	0.01M	0.603	0.187	0.803	0.397	0.439
		✓	0.03M	0.514	0.320	0.800	0.383	0.447
		✓	0.09M	0.596	0.220	0.782	0.397	0.433

[†] Independently implemented.

Table 7: Pretraining strategy ablation. All pretraining runs use 20 epochs. L2 and collision rate are averaged over 1, 2, and 3 seconds. #Params denotes the number of auxiliary parameters. Occ. and Act. denote occupancy and PAVER’s action conditioning.

Comparison of Pretraining Supervision

Table 7 separates the target from the budget. Detection-and-map pretraining reaches the lowest planning L2, but no task-supervised row matches the collision rate and NDS of sparse action targets. Raising the budget to 0.03M or 0.09M trades collision for L2, map mAP, or ADE, so a larger auxiliary head shifts the trade-off rather than improving transfer.

Feature Masking and Action State

Variants differ only in the ablated component: dropping the concatenated $[x, y, \psi, v]$ input, or sampling the original BEV features at the same action centers.

Table 8 shows that neither component helps safety alone: they leave collision at 0.520 and 0.550 against a 0.513 scratch baseline, and only together reach 0.187 with the lowest L2 and the highest NDS. Motion and map peak at the partial variants, so the pairing buys collision rather than uniform gains.

		Planning		Motion			Detection	Map
Mask	Action	L2 (m) ↓	Col. (%) ↓	ADE ↓	FDE ↓	MR ↓	NDS ↑	mAP ↑
		0.662	0.513	0.905	1.250	0.135	0.338	0.419
✓		0.655	0.520	0.808	1.084	0.115	0.396	0.443
	✓	0.705	0.550	0.797	1.080	0.123	0.388	0.445
✓	✓	0.603	0.187	0.803	1.086	0.121	0.397	0.439

Table 8: Component ablation. Mask denotes action-corridor feature masking with a shared learnable token, and Action denotes conditioning the prediction head on the ego-frame action state $[x, y, \psi, v]$, where ψ is the yaw heading. The variant using both is PAVER; the empty row is VAD-Tiny trained from scratch.

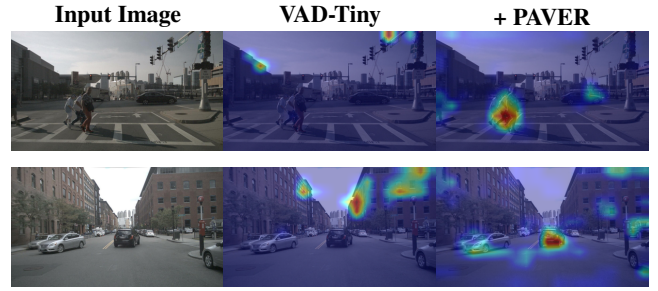


Figure 7: Grad-CAM results on nuScenes.

Method	Training Cost			Planning		Motion	Detection	Map
	PT	FT	Time	L2 (m) ↓	Col. (%) ↓	ADE ↓	NDS ↑	mAP ↑
VAD-Tiny [†]	48 [‡]	12	20.5h	0.701	0.730	0.840	0.363	0.413
VAD-Tiny [†]	0	60	21.3h	0.662	0.510	0.905	0.338	0.419
+ Ours	20	10	6.5h	<u>0.637</u>	0.300	0.851	0.357	0.361
+ Ours	20	20	10.1h	0.647	<u>0.220</u>	0.781	<u>0.387</u>	<u>0.425</u>
+ Ours	20	30	13.6h	0.603	0.187	<u>0.803</u>	0.397	0.439
VAD-Base	0	60	103h	0.719	0.217	<u>0.686</u>	0.460	<u>0.476</u>
+ Ours	20	10	31.8h	<u>0.646</u>	0.530	0.772	0.371	0.403
+ Ours	20	20	49.0h	0.757	<u>0.280</u>	0.698	0.442	0.460
+ Ours	20	30	66.1h	0.561	0.430	0.685	<u>0.446</u>	0.498

[†] Reproduced results.

[‡] Stage 1 pretrains detection, mapping, and motion prediction without planning.

Table 9: Training schedule ablation. Time denotes the total training time measured on 4 NVIDIA RTX 5090 GPUs. Bold: best; underlined: second-best.

Pretraining and Fine-Tuning Schedules

Table 9 shows that the pretraining stage pays for itself. Twenty pretraining epochs followed by 30 fine-tuning epochs cut total training time by about 36% on both models (21.3 to 13.6 hours for VAD-Tiny, 103 to 66.1 for VAD-Base) while giving the best L2 in each group. Metrics converge at different rates, so the shorter schedule is not uniformly better: ADE peaks at 20 fine-tuning epochs while L2, collision, NDS, and map mAP peak at 30, and on VAD-Base collision and NDS stay below the scratch baseline.

Conclusion

We presented PAVER, a lightweight pretraining method that organizes LiDAR-derived supervision into sparse action targets. Only the pretrained BEV encoder is transferred, leaving downstream camera-only inference unchanged. On VAD-Tiny, PAVER improves planning, collision, motion, and mapping metrics while preserving detection, supporting planning-aligned geometry as a useful initialization; broader architecture and closed-loop evaluation remain necessary.

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